

César Antonio Carvajal de la Haza

Mathematician · Statistician · Physicist

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PROFILE

Mathematics, Statistics and Physics graduate (GPA 8.0–8.3/10) with hands-on research experience in quantum many-body systems and entanglement entropy. Currently working as a Consulting Analyst specializing in process automation at Minsait (Indra Group), and starting an M.Sc. in Quantum Technologies (QTEP) in October 2026. Programming skills in Python, C/C++, and SQL, developed through research projects and algorithmic practice.

Full academic CV available at my personal space on the web: cesardelahaza.github.io.

EDUCATION

M.Sc. in Quantum Technologies (QTEP) — (Starting October 2026)

UIMP, Spain

Admitted to the QTEP Master's programme, ranked 2nd on the admissions list. Core focus on quantum computing, open quantum systems, and quantum sensors.

BSc Physics — 240 ECTS (2021 – 2025) | GPA: 8.1/10

UNED, Spain

Honours in Classical Mechanics, Vibrations & Waves, Quantum Physics II, Dynamical Systems. Final project: entanglement entropy and geometry in hard-core boson systems, using a Hubbard Hamiltonian and numerical diagonalisation in Python to test the entanglement Area Law (see Projects).

BSc Mathematics — 240 ECTS (2016 – 2021) | GPA: 8.0/10

University of Seville, Spain

Algebra, calculus, topology, cryptography, and computing. Languages: Haskell, Python, MATLAB, Maxima.

BSc Statistics — 240 ECTS (2016 – 2021) | GPA: 8.3/10

University of Seville, Spain

Statistical inference, multivariate analysis, ML, and computational statistics. Honours in Survey Design.

PROJECTS

MLWeather (2026)

Python, Scikit-learn, XGBoost, Keras, Streamlit · GitHub · Live app

End-to-end temperature forecasting pipeline for Seville: API data collection, preprocessing with lag features and cyclical encodings, comparison of 5 ML models (Linear Regression, XGBoost, Random Forest, SARIMA, LSTM), and interactive Streamlit app.

Entanglement Entropy and Geometry in Quantum Systems (2024 – 2025)

Final degree project — Physics · GitHub · Outreach article

Investigated entanglement structure in hard-core boson systems using a Hubbard Hamiltonian on graphs with varying topologies, testing the entanglement Area Law. Built the Entanglement Adjacency Matrix (EAM) to identify Area Law violations and entanglement monogamy effects; all simulations implemented in Python.

Algorithmic Problem Solving — LeetCode (2026 – present)

Self-directed · GitHub

Solving algorithmic (C++) and database (SQL: joins, aggregations) problems across all difficulty levels. Solutions version-controlled on GitHub.

Terminal & Shell Scripting (ongoing)

Self-directed

Command-line workflows in Unix/Linux: file manipulation, grep, and writing/executing .sh scripts to automate tasks.

TRAINING & CONTINUING EDUCATION

Introduction to Quantum Computing — 1.5 ECTS (March – April 2026)

UNED, Spain

GNSS Academy — Intensive Training Programme (January – July 2025)

6-month combined study and practical programme

Study: DGNSS, RTK, PPP, SBAS, GNSS data processing, SP3 orbit propagation, ionospheric/tropospheric corrections, advanced positioning. Project: GNSS+IMU sensor fusion simulator in Python/Linux with Least-Squares Estimation.

WORK EXPERIENCE

Consulting Analyst (Grade I) — RPA (June 2026 – present)

Minsait (Indra Group), Spain

Data, Intelligence & Robots unit, specialising in Robotic Process Automation (RPA). Completing UiPath Automation Developer Professional Training.

QA Automation Engineer (April – November 2022)

Deloitte, Spain

Automated test suites in Selenium/Java; co-implemented framework with Cucumber & Jenkins. Defect tracking in Jira; collaborative development with Git.

Private Tutor — Mathematics, Physics, Chemistry (several years)

Self-employed

Tutored students across multiple subjects; focus on clarity, reasoning, and problem-solving.

TECHNICAL SKILLS

Programming: Python, R, Haskell, MATLAB, SQL (intermediate), C/C++ (intermediate), Bash (basic)

Tools: Linux, Git, Jupyter, LaTeX, VSCode, gLAB

GNSS & Nav: DGNSS, RTK, PPP, SBAS, IMU fusion, SP3 propagation, ECEF transformations

Data & Maths: Statistical modelling, ML, optimisation, differential equations, cryptography

Quantum: Quantum mechanics, entanglement entropy, quantum computing (introductory)

LANGUAGES

Spanish (Native) · English (C1 — Proficient) · French (Basic, in progress)